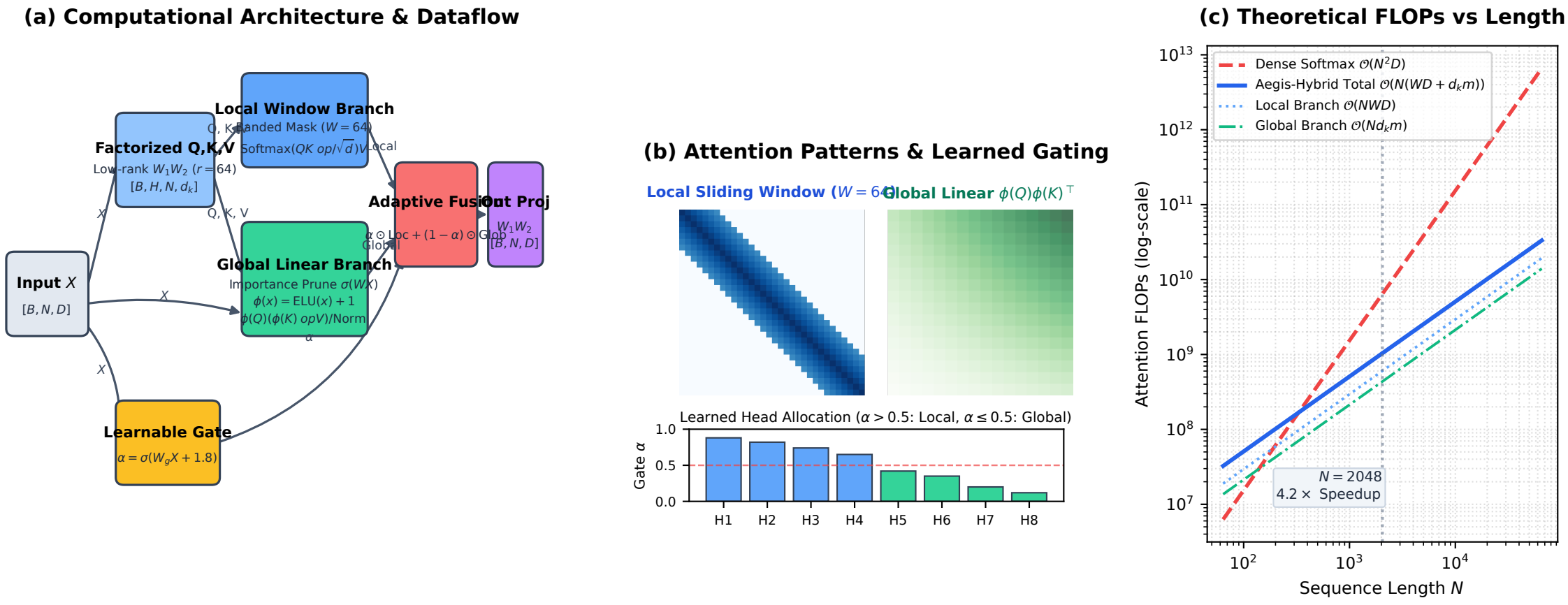

(a) Computational Architecture & Dataflow
Input X
$[B, N, D]$
Factorized Q,K,V
Low-rank W_1W_2 $(r = 64)$
$[B, H, N, d_k]$
Q, K, V
Local Window Branch
Q, K, V
Banded Mask $(W = 64)$
Softmax$(QK\ op/\sqrt{d})V$
Local
Global Linear Branch
Importance Prune $\sigma(WX)$
$\phi(x) = \mathrm{ELU}(x) + 1$
$\phi(Q)(\phi(K)\ opV)$/Norm
Global
Learnable Gate
$\alpha = \sigma(W_gX + 1.8)$
Adaptive Fusion
$\alpha \odot \mathrm{Loc} + (1 - \alpha) \odot \mathrm{Glob}$
Out Proj
W_1W_2
$[B, N, D]$
(b) Attention Patterns & Learned Gating
Local Sliding Window ($W = 64$)
Global Linear $\phi(Q)\phi(K)^\top$
Learned Head Allocation ($\alpha > 0.5$: Local, $\alpha \leq 0.5$: Global)
Gate α
1.0
0.5
0.0
H1
H2
H3
H4
H5
H6
H7
H8
(c) Theoretical FLOPs vs Length
Dense Softmax $\mathcal{O}(N^2D)$
Aegis-Hybrid Total $\mathcal{O}(N(WD + d_km))$
Local Branch $\mathcal{O}(NWD)$
Global Branch $\mathcal{O}(Nd_km)$
$N = 2048$
4.2 × Speedup
Attention FLOPs (log-scale)
10^{13}
10^{12}
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{2}
10^{3}
10^{4}
Sequence Length N